

Amir Choudhary

Greater Noida, India

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GitHub: [Amirchoudhary09](#) | LeetCode: [AMIR09](#) | GFG: [amirchoudharyb09](#) | Portfolio: codolio.com/profile/amirchoudharyb09

SUMMARY

Software Development Engineer skilled in Data Structures & Algorithms (DSA), system design, and full-stack development using MongoDB, Express.js, React.js, and Node.js (MERN stack). Built scalable, real-time web applications with optimized APIs, secure authentication, and AI/ML integrations. Strong experience in low-level GPU programming using C++ and DirectX 12, implementing efficient rendering pipelines.

EDUCATION

G.L. Bajaj Institute of Technology and Management

Greater Noida, India

B.Tech in Computer Science – Artificial Intelligence and Machine Learning (AIML) — 8.01/10 CGPA

2022 – 2026

EXPERIENCE

Software Development Engineer (SDE)

April 2026 – Present

Wasp3D

Noida, India (Onsite)

- Engineered a low-level graphics rendering system using **C++ and DirectX 12**, implementing the complete rendering pipeline with GPU resource management (Vertex/Index Buffers, Descriptor Heaps, RTV).
- Designed and optimized **Root Signature and Pipeline State Objects (PSO)** to reduce runtime overhead and improve pipeline efficiency.
- Implemented modular architecture using **OOP principles** for device initialization, command queue, and resource handling, reducing code coupling across rendering modules.
- Managed **Resource Barriers, GPU synchronization**, and integrated **HLSL shaders (Vertex & Pixel)** ensuring correct state transitions, smooth frame execution, and efficient CPU-GPU data flow.

Software Development Engineer Intern (SDE Intern)

March 2025 – May 2025

Bluestocks Fintech

Remote

- Developed a responsive **Services** page using **React.js and Tailwind CSS** with modular component architecture, improving page load performance by **20%**; built and integrated **full CRUD REST APIs** using **Express.js and MongoDB** for service management.
- Tested APIs using **Postman** with proper error handling, status codes, and edge case validation, ensuring smooth frontend-backend data flow.
- Enhanced UI consistency across modules and collaborated in **Agile sprints**, presenting features to stakeholders in bi-weekly demos.

PROJECTS

AshGuard – Women Safety and Travel Assistant | *React.js, Python, Express.js, Gemini AI, ML* | [Live](#)

[GitHub](#)

- Developed an **AI-powered women safety and travel web app** with real-time SOS alerts, live location tracking, emergency response, and **1-Tap SOS system** (fake call, live recording, guardian alert, police dial, siren).
- Implemented a **Safe Route Finder** using **Google Maps API** and **K-Means Clustering** to detect crime-prone areas and suggest secure routes, achieving **93% route accuracy**.
- Built full-stack system using **React.js, Tailwind CSS, Python, Express.js** with **Gemini AI chatbot** and **JWT authentication**, deployed on **Vercel & Railway**, serving **200+ active users**.

Lost and Found Community Platform | *MERN Stack – React.js, Node.js, Express.js, MongoDB* | [Live](#)

[GitHub](#)

- Built a full-stack lost & found web application with image uploads, location tagging, AI-powered filtering, and a matching algorithm to suggest potential item matches.
- Developed **RESTful APIs** using **Node.js and Express.js** with **MongoDB Atlas**, implementing **JWT and OAuth 2.0 (Google Sign-In)** for secure authentication and seamless onboarding.
- Created an interactive **React.js** frontend with real-time filtering by location and category; optimized database queries achieving **40% faster search response**.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, C++, SQL

Frontend: React.js, Next.js, Redux, Tailwind CSS, React Query

Backend: Node.js, Express.js, RESTful APIs, JWT, OAuth 2.0, Mongoose, WebSockets, Zod

Databases: MongoDB (Atlas), MySQL, Redis

Generative AI: Gemini API, OpenAI API, LangChain, Prompt Engineering, Retrieval-Augmented Generation (RAG)

Tools & Platforms: Git, GitHub, Postman, VS Code, Vercel, Railway

Core Concepts: Data Structures & Algorithms, OOP, DBMS, System Design, Operating Systems, Computer Networks, SDLC

ACHIEVEMENTS

Competitive Programming: Solved **900+ DSA problems** on [LeetCode](#) (Max Rating: **1600**) and [GeeksforGeeks](#).

Top 3 at Adobe Emerge Hackathon 2025 – IIT Delhi (National-level competition)

Top 6 at Hack For Impact 2025 – IIIT Delhi ([Certificate](#))

Top 10 at HackSprint Hackathon 2024 – GDG GL Bajaj ([Certificate](#))

CERTIFICATIONS & VIRTUAL INTERNSHIPS

Google for Developers Virtual Internship | [Google](#)

June 2024

Web Development Internship | [See Caller](#)

Jan 2025

CCNA / Python Certification | [Cisco](#)

May 2025

Coding Competition Certificate | [HackSprint](#), GDG GL Bajaj

Nov 2024